

QUESTION 1 (Start a new page)

- (a) Express $0.373737\ldots$ as a proper fraction in simplest form.
- (b) Find the size of the acute angle between the lines $2x + y = 5$ and $3x - y = 1$.
- (c) Find all solutions to : $\frac{1}{x-2} \leq 4$.
- (d) Differentiate with respect to x : $y = \tan^{-1} 2x$.
- (e) Solve $2 \cos^2 x + 3 \sin x - 3 = 0$, for $0 \leq x \leq 2\pi$.

QUESTION 2 (Start a new page)

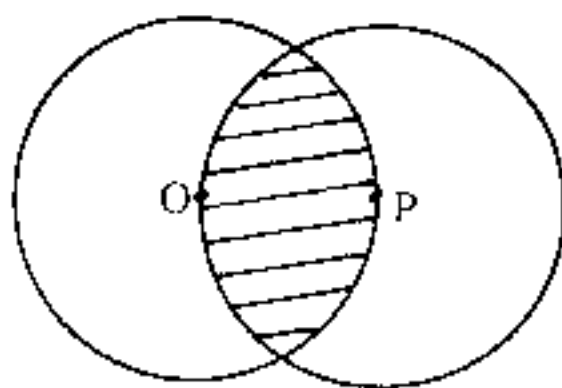
- (a) Prove the identity: $\frac{\sin 2\theta}{1 + \cos 2\theta} = \tan \theta$.
- (b) Given that $\frac{dy}{dx} = \frac{1}{1+x^2}$, and $x=1$ when $y=0$, find y when $x = \sqrt{3}$.
- (c) Evaluate : $\int_0^{\sqrt{3}} \frac{dx}{\sqrt{3-x^2}}$
- (d) Prove by mathematical induction that $3^{4n} - 1$ is divisible by 80 for all positive integers n .

QUESTION 3 (Start a new page)

- (a) Let the equation of motion of an object moving x metres along a straight line after t seconds be: $x(t) = 4\sin 3t - 5\cos 3t$ ($t \geq 0$). Show that its motion is Simple Harmonic, and find its period of motion.

- (b) Evaluate : $\int_0^{\frac{\pi}{8}} \cos^2 2x dx$

- (c) In the diagram shown, the two circles are of radius 1 metre and pass through centres O and P . Find the area of their intersection (to two decimal places).



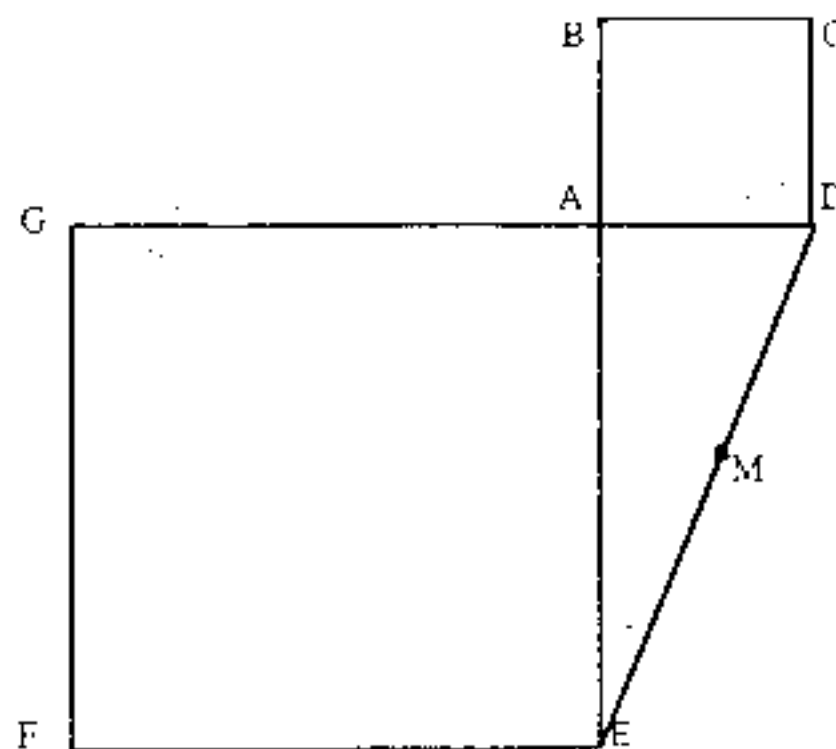
QUESTION 4 (Start a new page)

- (a) If 5% of monkeys are colourblind, what is the probability that a random sample of 20 monkeys should contain at least two colourblind monkeys? (Answer to three decimal places.)
- (b) A person invests \$1000 at the beginning of each year in a superannuation fund. If interest is paid at 9% per annum, find :
- the value of the investment at the end of 30 years.
 - how many years would elapse for the investment to be worth \$50,000.
- (c) Neatly sketch $y = 3\cos^{-1}\pi x$, and state its domain and range.

QUESTION 5 (Start a new page)

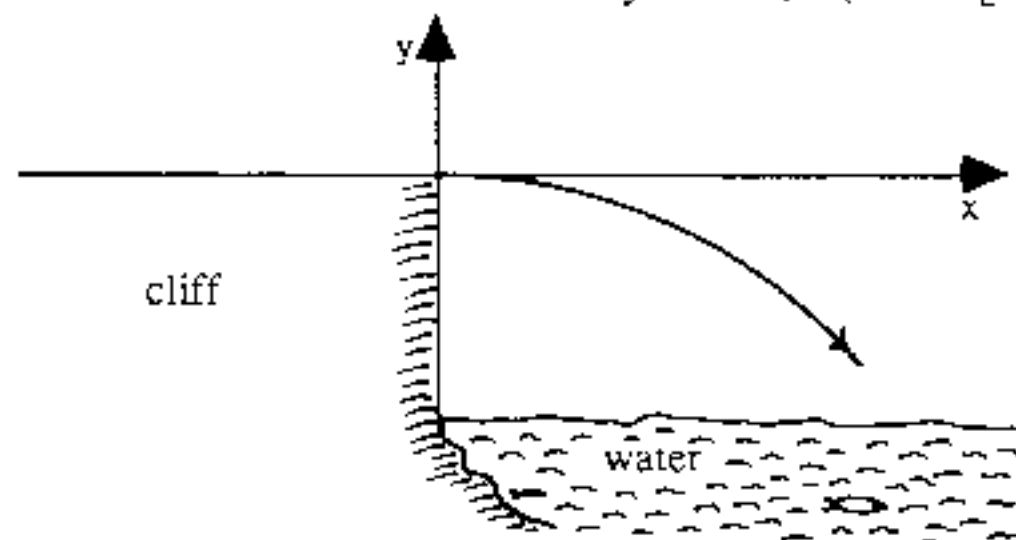
- (a) The acceleration of a particle is given by $\frac{d^2x}{dt^2} = 16(1+x)$, where x cm. is the displacement from the origin. When $t = 0$, $x = 0$ and $v = 4$ cm./sec.
- Derive an expression for its velocity in terms of its displacement.
 - Deduce that its displacement function is $x(t) = e^{4t} - 1$.
- (b) Evaluate $\cot 2\theta$ if $\cot^2\theta - \cot\theta = 1$.
- (c) ABCD and AEFG are two squares of different areas, and $GD \perp BE$. M is the mid point of DE.
- Give a reason why DE is the diameter of the circle with points A, D and E on its circumference.
 - Prove that BDEG is a cyclic quadrilateral.
 - Prove that $AM \perp BG$.

Copy the diagram below onto your answer sheet.



QUESTION 6 (Start a new page)

- (a) An object is projected horizontally from the top edge of a vertical cliff 40 metres above sea level with a velocity of 40m/s. (Take $g = 10 \text{ m/s}^2$)



- (i) Using the top edge of the cliff as origin, prove that the parametric equations of the path of the object are:
- $$x = 40t \quad y = -5t^2$$
- (ii) Calculate when and where the object hits the water.
- (iii) Find the velocity (magnitude and direction) of the object the instant it hits the water.
- (b) The inside of a vessel used for water has the shape of a solid of revolution obtained by the rotation of the parabola $9y = 8x^2$ about the y -axis. The depth of the vessel is 8 cm.
- (i) Prove that a volume of water h cm. from its bottom is $\frac{9}{16} \pi h^2$.
- (ii) If water is poured into the vessel at a rate of $20 \text{ cm}^3/\text{sec.}$, find the rate at which the level of water is rising when the vessel is half full.

QUESTION 7 (Start a new page)

- (a) Two parametric points $P(2p, p^2)$ and $Q(2q, q^2)$ lie on the parabola $x^2 = 4y$, and the line through PQ is parallel to the line $y = mx$.
- (i) Show that $p + q = 2m$.
- (ii) Derive the equation of the normal to the parabola at the point P .
- (iii) Find the co-ordinates of N , the point of intersection of the normals from P and Q .
- (iv) Determine the locus of N as the line PQ moves parallel to the line $y = mx$. State any restrictions on the locus of N .
- (b) A_n and B_n are two series given by :

$$A_n = 1^2 + 5^2 + 9^2 + 13^2 + \dots + (4n - 3)^2$$

$$B_n = 3^2 + 7^2 + 11^2 + 15^2 + \dots \quad \text{for } n = 1, 2, 3, \dots$$

- (i) Find the n th term of B_n .
- (ii) If $S_{2n} = A_n - B_n$, prove that $S_{2n} = -8n^2$.
- (iii) Hence, or otherwise, evaluate :
- $$101^2 - 103^2 + 105^2 - 107^2 + \dots + 1993^2 - 1995^2.$$

END OF PAPER

$$1 (a) \frac{37}{49}$$

$$(b) \frac{\pi}{4}$$

$$(c) x < 2, x \geq 9/4$$

$$(d) \frac{2}{1+4x^2}$$

$$(e) \frac{\pi}{6}, \frac{5\pi}{6}, \frac{\pi}{2}$$

$$2 (a) -$$

$$(b) \frac{\pi}{12}$$

$$(c) \frac{\pi}{2}$$

$$(d) -$$

$$3 (a) \tau = \frac{2\pi}{3}$$

$$(b) \frac{\pi}{16} + \frac{1}{8}$$

$$(c) 1, 2, 3 \quad (4D)$$

$$(d) 0.264 \quad (3D)$$

$$(e) (i) \$148,575$$

$$(ii) 19 \quad (\text{nearest yr})$$

$$(c) D: \frac{1}{\sqrt{e}} \leq x \leq \frac{1}{e}$$

$$R: 0 \leq y \leq 3\pi$$

$$7 (a) (i) v = 4(1+x)$$

$$(ii) -$$

$$(b) \frac{1}{2}$$

$$(c) -$$

$$6 (a) (i) -$$

$$(ii) 80\sqrt{2}, t = 2\sqrt{2} \text{ sec}$$

$$(iii) 35^\circ \text{ with speed } 20\sqrt{6} \text{ m/s}$$

$$(b) \frac{20\sqrt{2}}{40} \text{ amp}$$

$$7 (a) (i) -$$

$$(ii) x + py = p^2 + 2p$$

$$(iii) N[-pq(p+q), p^2q + pq^2 + 2]$$

$$(iv) y = \frac{x}{2m} + 4m^2 + 2$$

$$\text{Restrictions } y > 3m^2 + 2$$

$$x > -2m^2$$

$$(b) (i) T_n = (4n-1)^2$$

$$(ii) -$$

$$(iii) -$$